

CHAPTER 1: PREREQUISITES FOR CALCULUS

Definition

Inequalities

Def: Inequality is a sentence built from expressions using one or more the symbols $<$, $>$, $\leq$ or $\geq$ for example $x + y < 1$, $-1 \leq 3x \leq 2$

Properties of inequalities:Let $a, b, c, d \in \mathbb{R}$

- 1) If $a < b$ and $c < d$ then $a + c < b + d$.
- 2) If $a < b$ then $a \pm c < b \pm c$.
- 3) If $a < b$ and $c \in \mathbb{R}^+$ then $ac < bc$ and $a/c < b/c$.
- 4) If $a < b$ and $c \in \mathbb{R}^-$ then $ac > bc$ and $a/c > b/c$.
- 5) If $a < b$ and $a, b \in \mathbb{R}^+$ or $a, b \in \mathbb{R}^-$ then $1/a > 1/b$.

Types of intervals: الفترات

Notation	Set description	Type
(a, b)	$\{x: a < x < b\}$	Open
$[a, b]$	$\{x: a \leq x \leq b\}$	Closed
$[a, b)$	$\{x: a \leq x < b\}$	Half open or Half closed
$(a, b]$	$\{x: a < x \leq b\}$	Half open or Half closed
(a, ∞)	$\{x: x > a\}$	Open
$[a, \infty)$	$\{x: x \geq a\}$	Closed
$(-\infty, b)$	$\{x: x < b\}$	Open
$(-\infty, b]$	$\{x: x \leq b\}$	Closed

Example: Solve the following inequalities and show their solution sets on the real line

1) $2x - 1 < x + 3$, 2) $-3 \leq 6x - 1 < 3$,

3) $\frac{6}{x-1} \geq 5$, 4) $x^2 - 7x + 10 > 0$

Solution:

1) $2x - 1 < x + 3 \Rightarrow 2x - x < 3 + 1 \Rightarrow x < 4$

Then the set of solution is $(-\infty, 4)$

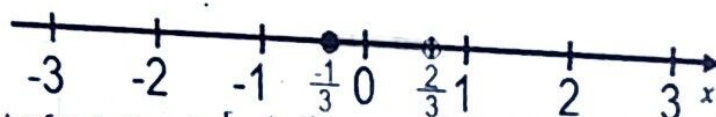


$$2) -3 \leq 6x - 1 < 3 \Rightarrow -3 + 1 \leq 6x - 1 + 1 < 3 + 1$$

$$\Rightarrow -2 \leq 6x < 4$$

$$\Rightarrow \frac{-2}{6} \leq x < \frac{4}{6}$$

$$\Rightarrow \frac{-1}{3} \leq x < \frac{2}{3}$$



Then the set of solution is $[-\frac{1}{3}, \frac{2}{3})$

$$\times 3) \frac{6}{x-1} \geq 5 \Rightarrow \frac{6}{x-1} - 5 \geq 0 \Rightarrow \frac{-5x+11}{x-1} \geq 0$$

$$\text{First let } -5x + 11 = 0 \Rightarrow x = \frac{11}{5} \Rightarrow x = 2\frac{1}{5}$$

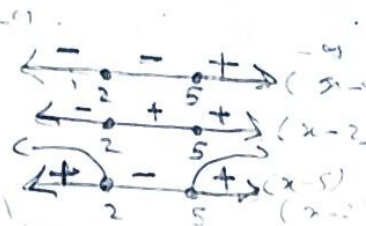
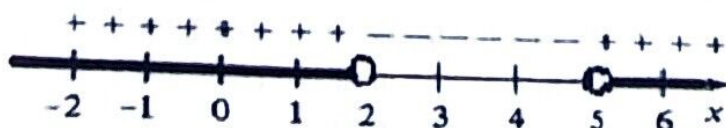
$$\text{And let } x - 1 = 0 \Rightarrow x = 1$$



Then the set of solution is $(1, \frac{11}{5}]$

$$4) x^2 - 7x + 10 > 0 \Rightarrow (x-5)(x-2) > 0$$

$$\text{Let } x - 5 = 0 \Rightarrow x = 5 \text{ and let } x - 2 = 0 \Rightarrow x = 2$$



Then the set of solution is $(-\infty, 2) \cup (5, \infty)$

Absolute value

The absolute value of a number x , denoted by $|x|$, is

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

For example: 1) $|3| = 3$, 2) $|-3| = 3$, 3) $|0| = 0$,
4) $|\sqrt{2} - 2| = 2 - \sqrt{2}$ because $2 > \sqrt{2}$

Properties of absolute value:

Let $a, b \in \mathbb{R}$ then

$$1) |-a| = a, \quad a \in \mathbb{R}^+$$

$$3) |a \cdot b| = |a| \cdot |b|$$

$$5) \text{ If } |a| = b (b \geq 0) \Rightarrow a = b \text{ or } a = -b$$

$$7) \text{ If } |a| = 0 \Rightarrow a = 0$$

$$2) \sqrt{a^2} = |a|$$

$$4) \left| \frac{a}{b} \right| = \frac{|a|}{|b|} \quad (b \neq 0)$$

$$6) \text{ if } |a| = |b| \Rightarrow a = b \text{ or } a = -b$$

Example: Solve the following equations

$$1) |3x + 12| + 7 = 7, \quad 2) |x - 3| - 2 = -5, \quad 3) |x + 5| = |2x - 1|$$

Solution:

$$\begin{aligned} 1) |3x + 12| + 7 = 7 &\Rightarrow |3x + 12| = 0 \\ &\Rightarrow 3x + 12 = 0 \\ &\Rightarrow x = -4 \end{aligned}$$

Then the set of solutions is $\{-4\}$.

$$2) |x - 3| - 2 = -5 \Rightarrow |x - 3| = -3$$

Since -3 is a negative number then the equation has not solution.

$$\begin{aligned} 3) |x + 5| = |2x - 1| &\Rightarrow x + 5 = 2x - 1 \quad \text{or } x + 5 = -(2x - 1) \\ &\Rightarrow x - 2x = -1 - 5 \quad \text{or } x + 5 = -2x + 1 \\ &\Rightarrow -x = -6 \quad \text{or } x + 2x = -5 + 1 \\ &\Rightarrow x = 6 \quad \text{or } 3x = -4 \\ &\Rightarrow x = 6 \quad \text{or } x = -\frac{4}{3} \end{aligned}$$

Then the set of solution is $\left\{6, -\frac{4}{3}\right\}$

Remark: Let $a \in \mathbb{R}^+$ then

$$1) |x| \leq a \iff (\text{if and only if}) \quad -a \leq x \leq a$$

$$2) |x| \geq a \iff (\text{if and only if}) \quad x \leq -a \quad \text{or} \quad x \geq a$$

Example: Solve each inequality

1) $|x - 5| \leq 2$, 2) $|2x + 1| > 17$, 3) $|x + 4| < -2$, 4) $|x + 1| \geq -1$

5) $|2x + 6| \leq 0$, 6) $|x + 1| < 0$

Solution:

1) $|x - 5| \leq 2 \Rightarrow -2 \leq x - 5 \leq 2$
 $\Rightarrow 3 \leq x \leq 7$

Then the set of solution is $[3, 7]$

2) $|2x + 1| > 17 \Rightarrow 2x + 1 > 17 \text{ or } 2x + 1 < -17 \xrightarrow{-1} 2x > 16 \text{ or } 2x < -18$
 $\xrightarrow{\div 2} x > 8 \text{ or } x < -9$

Then the set of solution is $(8, \infty) \cup (-\infty, -9)$ 3) $|x + 4| < -2$ Since -2 is a negative number and $| \cdot | \geq 0$ then the inequality has not solution.

4) $|x + 1| \geq -1 \Rightarrow |x + 1| \geq 0 \Rightarrow x + 1 \geq 0 \text{ or } x + 1 \leq 0 \xrightarrow{-1} x \geq -1 \text{ or } x \leq -1$

Then the set of solution is $(-\infty, -1] \cup [-1, \infty) = \mathbb{R}$

5) $|2x + 6| \leq 0 \Rightarrow |2x + 6| = 0 \Rightarrow 2x + 6 = 0 \xrightarrow{-6} 2x = -6 \xrightarrow{\div 2} x = -3$

Then the set of solution is $\{-3\}$ 6) $|x + 1| < 0$ Since $| \cdot | \geq 0$ then the inequality has not solution.